

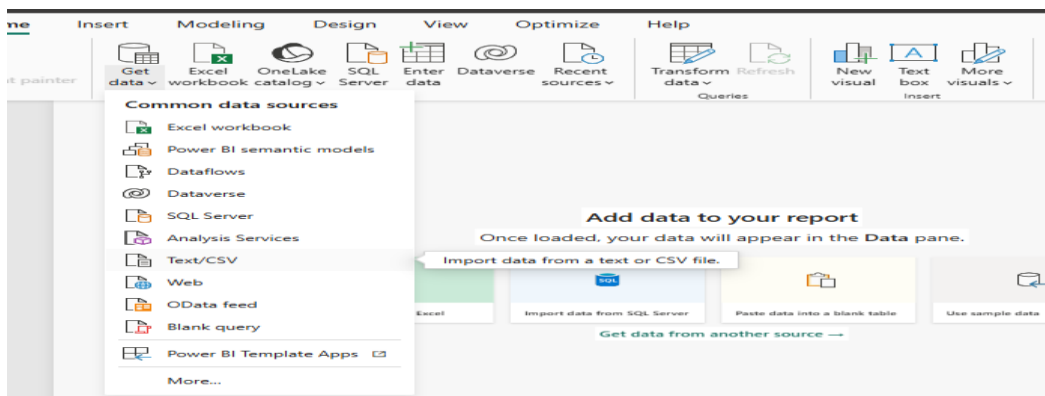
Practical No. 2

Aim: Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sqlserver / Power BI.

The three stages of ETL are:

1. Extraction
Data is collected from one or more data sources such as Excel files, CSV files, databases or APIs.
2. Transformation
The extracted data is cleaned and modified according to analytical requirements. Common transformations include:
 - Removing duplicate records
 - Handling missing values
 - Changing data types
 - Renaming columns
 - Filtering records
 - Creating calculated columns
 - Categorizing data
3. Loading
The transformed data is loaded into the target system or data model where it can be analyzed and visualized.

Step 1: Import the required CSV datasets into Power BI Desktop.



Step 2: Import the Products dataset and preview the product data.

The screenshot shows the 'Import data from a text or CSV file' dialog box in Power BI Desktop. The 'File Origin' is set to 'Text/CSV'. The 'Delimiter' is set to 'Comma'. The 'Data Type Detection' is set to 'Based on first 200 rows'. The 'Table Name' is 'products'. The 'Table Structure' is shown as follows:

ProductID	ProductName	SupplierID	CategoryID	QuantityPerUnit	UnitPrice	UnitsInStock	UnitsOnOrder	ReorderLevel
1	Chai	1	1	10 boxes x 20 bags	18	39	0	0
2	Chang	1	1	24 - 12 oz bottles	19	17	40	0
3	Aniseed Syrup	1	2	12 - 550 ml bottles	10	13	70	0
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22	53	0	0
5	Chef Anton's Gumbo Mix	2	2	36 boxes	21	25	0	0
6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25	120	0	0
7	Uncle Bob's Organic Dried Pears	3	2	12 - 1 lb pkgs.	30	15	0	0
8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars	40	6	0	0
9	Mishi Kobe Niku	4	6	18 - 500 g pkgs.	97	29	0	0
10	Ikura	4	6	12 - 200 ml jars	31	31	0	0
11	Queso Caboties	5	4	1 kg pkg.	21	22	30	0
12	Queso Manchego La Pastora	5	4	10 - 500 g pkgs.	38	86	0	0
13	Konbu	6	6	2 kg box	6	24	0	0
14	Tofu	6	7	40 - 100 g pkgs.	23.25	35	0	0
15	Genen Shoyu	6	2	24 - 250 ml bottles	15.5	39	0	0
16	Pavlova	7	3	32 - 500 g boxes	17.45	29	0	0
17	Alice Mutton	7	6	20 - 1 kg tins	39	0	0	0
18	Carnarvon Tigers	7	6	16 kg pkg.	62.5	42	0	0
19	Teatime Chocolate Biscuits	8	3	10 boxes x 12 pieces	9.2	25	0	0
20	Sir Rodney's Marmalade	8	3	30 gifts boxes	81	40	0	0

Step 3: Retain the required Products columns: ProductID, ProductName, QuantityPerUnit and UnitsInStock.

The screenshot shows the Power Query Editor with the 'products' table loaded. The table has the following columns: ProductID, ProductName, SupplierID, CategoryID, QuantityPerUnit, UnitPrice, UnitsInStock, UnitsOnOrder, and ReorderLevel. The 'ProductID' column is highlighted, and the 'Table.TransformColumnTypes' function is applied to the table. The function is defined as: Table.TransformColumnTypes(#"Promoted Headers",{{{"ProductID", Int64.Type}, {"ProductName", type text}, {"SupplierID", type text}, {"CategoryID", Int64.Type}, {"QuantityPerUnit", type text}, {"UnitPrice", Int64.Type}, {"UnitsInStock", Int64.Type}, {"UnitsOnOrder", Int64.Type}, {"ReorderLevel", Int64.Type}}}). The 'ProductID' column is set to 'Int64.Type', 'ProductName' is set to 'type text', 'SupplierID' is set to 'type text', 'CategoryID' is set to 'Int64.Type', 'QuantityPerUnit' is set to 'type text', 'UnitPrice' is set to 'Int64.Type', 'UnitsInStock' is set to 'Int64.Type', 'UnitsOnOrder' is set to 'Int64.Type', and 'ReorderLevel' is set to 'Int64.Type'. The 'Table.TransformColumnTypes' function is applied to the table, and the resulting table is shown below:

ProductID	ProductName	SupplierID	CategoryID	QuantityPerUnit	UnitPrice	UnitsInStock	UnitsOnOrder	ReorderLevel
1	Chai	1	1	10 boxes x 20 bags	18	39	0	0
2	Chang	1	1	24 - 12 oz bottles	19	17	40	0
3	Aniseed Syrup	1	2	12 - 550 ml bottles	10	13	70	0
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22	53	0	0
5	Chef Anton's Gumbo Mix	2	2	36 boxes	21	25	0	0
6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25	120	0	0
7	Uncle Bob's Organic Dried Pears	3	2	12 - 1 lb pkgs.	30	15	0	0
8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars	40	6	0	0
9	Mishi Kobe Niku	4	6	18 - 500 g pkgs.	97	29	0	0

Step 4: Change the data type of UnitsInStock to Whole Number.

Products table data:

ProductID	ProductName	QuantityPerUnit	UnitsInStock
10	boxes x 20 bags		39
11	24 - 12 oz bottles		17
12	12 - 550 ml bottles		13
13	48 - 6 oz jars		53
14	36 boxes		0
15	12 - 6 oz jars		120
16	12 - 1 lb pkgs.		15
17	12 - 12 oz jars		6
18	18 - 500 g pkgs.		29
19	12 - 200 ml jars		31

Step 5: Import the Orders dataset into Power BI.

Orders table data:

OrderID	ProductID
1	10248
2	10248
3	10248
4	10249
5	10249

Step 6: Import the Order_Details dataset into Power BI.

Order_Details table data:

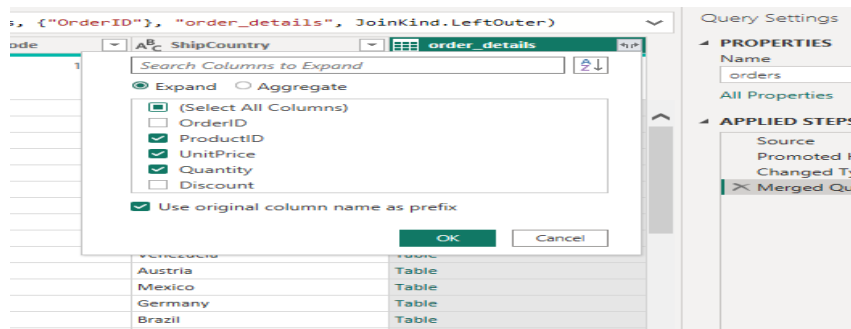
OrderID	ProductID	Customer
1	10248	VINET
2	10249	TOMSP
3	10250	HANAR
4	10251	VICTE
5	10252	SUPRD
6	10253	HANAR
7	10254	CHOPS

Step 7: Merge the Orders and Order_Details queries using the common OrderID field.

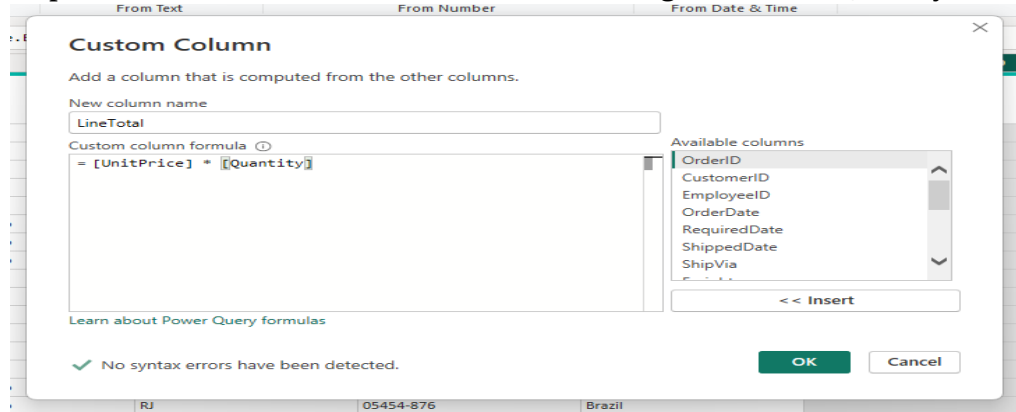
Merge dialog box details:

- Join Kind: Left Outer (all from first, matching from second)
- Use fuzzy matching to perform the merge: ☐
- Fuzzy matching options: ☐
- The selection matches 830 of 830 rows from the first table.

Step 8: Expand Order_Details and select ProductID, UnitPrice and Quantity.



Step 9: Create the LineTotal calculated column using $\text{UnitPrice} \times \text{Quantity}$.



Step 10: Verify the transformed data with the calculated LineTotal column.

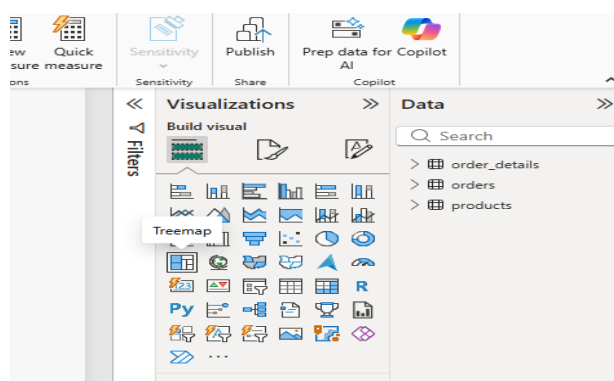
Parse ▾ Statistics Standard Scientific Information ▾ Date Time Duration ▾

From Text From Number From Date & Time

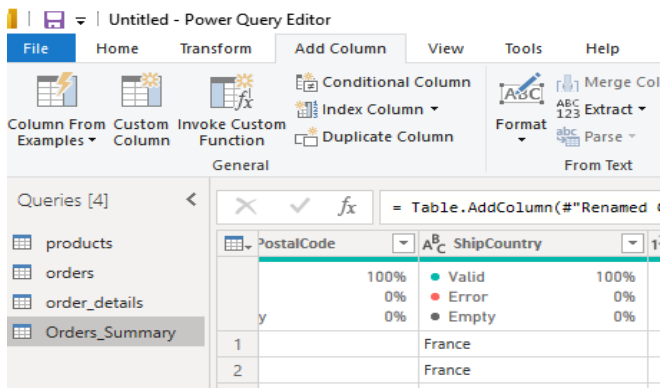
#"Renamed Columns", "LineTotal", each [UnitPrice] * [Quantity])

	123 ProductID	1.2 UnitPrice	123 Quantity	ABC 123 LineTotal
100%	Valid	100%	Valid	100%
0%	Error	0%	Error	0%
0%	Empty	0%	Empty	0%
	11	14	12	168
	42	9.8	10	98
	72	34.8	5	174
	14	18.6	9	167.4
	51	42.4	40	1696
	41	7.7	10	77
	51	42.4	35	1484
	65	16.8	15	252
	22	16.8	6	100.8
	57	15.6	15	234
	65	16.8	20	336
	20	64.8	40	2592

Step 11: Verify that the Orders, Order_Details and Products tables are loaded successfully in Power BI Data view.



Step 12: Create and display the Orders_Summary query in Power Query.



Close	New Query	Data Sources	Parameters	Output Data	Query	Man
Queries [4]						
products	orders	order_details	Orders_Summary			
= Table.Distinct(#"Removed Other Columns", {"OrderID"})						
1	2	3	4	5	6	7
OrderID	OrderDate	Valid	Error	Empty	Valid	Error
10248	04-07-1996 00:00:00	100%	0%	0%	100%	0%
10249	05-07-1996 00:00:00	100%	0%	0%	100%	0%
10250	08-07-1996 00:00:00	100%	0%	0%	100%	0%
10251	08-07-1996 00:00:00	100%	0%	0%	100%	0%
10252	09-07-1996 00:00:00	100%	0%	0%	100%	0%
10253	10-07-1996 00:00:00	100%	0%	0%	100%	0%
10254	11-07-1996 00:00:00	100%	0%	0%	100%	0%
10255	12-07-1996 00:00:00	100%	0%	0%	100%	0%
10256	15-07-1996 00:00:00	100%	0%	0%	100%	0%
10257	16-07-1996 00:00:00	100%	0%	0%	100%	0%
10258	17-07-1996 00:00:00	100%	0%	0%	100%	0%
10259	18-07-1996 00:00:00	100%	0%	0%	100%	0%
10260	19-07-1996 00:00:00	100%	0%	0%	100%	0%
10261	19-07-1996 00:00:00	100%	0%	0%	100%	0%
10262	22-07-1996 00:00:00	100%	0%	0%	100%	0%
10263	23-07-1996 00:00:00	100%	0%	0%	100%	0%
10264	24-07-1996 00:00:00	100%	0%	0%	100%	0%
10265	25-07-1996 00:00:00	100%	0%	0%	100%	0%

Step 14: View the final Power BI data model and verify the relationships between the tables.

